

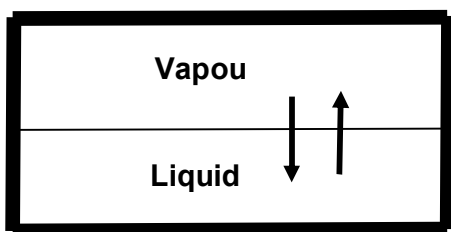
CHE 2201 – Physical Chemistry II

Phase Equilibria

Lec. 1. Terminology. Phase Rule. Thermodynamics of Phase Changes in Univariant Systems

1.1. Introduction and Terminology

Let's say that we have a sample of water in a closed evacuated vessel at a constant temperature. Some water molecules will go from water vapour, and these molecules may enter the liquid phase by collisions with the water surface. After some time, an equilibrium state will be reached, where the number of H_2O molecules leaving liquid water is equal to the number of H_2O molecules in the vapour state entering the liquid phase.



In this example, if we consider only the liquid, we will see that it has a definite boundary and the distribution of molecules throughout it is uniform or homogeneous. Such part of a system, which has a definite boundary and homogeneous distribution of molecules throughout is called a phase. Similarly, water molecules in the gaseous state represent another phase, and it is the vapour phase. So, in this case we see that two phases, one liquid and the other – gaseous, are in equilibrium. Such an equilibrium is called a phase equilibrium.

More precisely, we can define a phase, as the totality of all the parts of a system identical at all points in composition and in all chemical and physical properties, and separated from other parts of the system by a visible dividing surface (i.e. interface).

For example, in a system consisting of water and ice, all the pieces of ice form one phase, and water – another phase. This definition of phase is true on condition that external

fields are absent or their intensity is the same at all points of the system. Further, each phase of a system is characterised by its own equation of state (i.e. an equation of the type $f(p, V, T, \dots) = 0$).

If we consider a system in thermodynamic equilibrium, it will show the following features.

- I. The system is in mechanical equilibrium. It means that the resultant of all the mechanical forces in the system is zero (i.e. one part of the system does not move with respect to the other). The condition of mechanical equilibrium of a system can be specified by giving the pressure (p) of the system.
- II. The system is in thermal equilibrium. By this we mean that the temperature (T) throughout the system is constant (i.e. various parts of the system will have the same temperature). So, the condition of thermal equilibrium of a system can be specified by giving the temperature (T) of the system.
- III. The system is in diffusional equilibrium, which means that the chemical composition of the system is uniform throughout. The condition of diffusional equilibrium of a system, having C number of components is given by specifying $(C - 1)$ composition variables. By doing so, we specify the composition of the entire system.

So, now we have to define the terms, number of components and the composition variable. Every substance that can be separated from a system and exist outside it is called a constituent of the system. For example, in an aqueous solution of NaCl, the constituents are H_2O and NaCl. The ions Na^+ and Cl^- cannot be included among the constituents, although they exist in the solution in the form of freely moving, independent particles.

The amount of each of the constituents of a system, in which there are no chemical reactions, does not depend on the amount of other substances. If chemical reactions take place in an equilibrium system, the amounts of its constituents depend on one another, and the composition of its phases can be determined if the concentrations of part of the constituents are known.

Constituents whose concentrations determine the composition of the phases of a given equilibrium system are called the independent constituents or the components of the system. Any constituent may be selected as a component.

As we will see later, the properties of an equilibrium system are determined not by what constituents have been selected as its components, but by their number, i.e. by the number of components (C). The number of components is either equal to the number of constituents (in the absence of chemical reactions) or is less than it.

The number of components is equal to the number of constituents of the system minus the number of equations relating the concentrations of these substances in an equilibrium system. Here, the concentration means the amount in moles per unit volume or the partial pressure or the chemical potential (μ). A different definition which reflects the same idea also possible, namely, the number of components of a system is the smallest number of constituents sufficient for determining the composition of any phase of the system.

For example, let us consider a solution of NaCl. In this system, we can change the amount of NaCl without changing the amount of H₂O, simply by adding some NaCl. It means that there are two independent constituents in this solution, or in other words, number of components of the system equals to 2.

So, Number of Components (C) = 2; and

Number of Composition Variables = $C - 1 = 2 - 1 = 1$.

This means that we can specify the composition of the NaCl solution by giving just one ($C - 1 = 1$) composition variable, i.e. the concentration. For example, when we say 1 mol/dm³ NaCl solution, it means that there is 1 mole of NaCl in 1 dm³ of water.

Now, let us consider a solution containing NaCl, KCl and H₂O. Here, each of the 3 constituents can be varied without affecting others, and as such, the number of components (C) is equal to 3. To specify the composition of this system, we have to give ($C - 1$) = (3 - 1) = 2, composition variables, namely the concentrations of NaCl and KCl.

Let us now consider a little more complicated system of acetic acid in water. As we know, in this case, the following equilibrium exists.



Here, although there are four chemical species in the system, we can vary only CH_3COOH and H_2O independently. The amounts of other two species (H_3O^+ and CH_3COO^-) will change automatically depending on the value of the equilibrium constant (K).

$$K = [\text{C}_{\text{H}_3\text{O}^+} \times \text{C}_{\text{CH}_3\text{COO}^-}] / [\text{C}_{\text{CH}_3\text{COOH}} \times \text{C}_{\text{H}_2\text{O}}] \quad (1.2)$$

So, the number of components in this system is two ($C = 2$). To specify the composition of the system, we have to give $(C - 1) = 2 - 1 = 1$, composition variable, namely the concentration of CH_3COOH .

So, when a system is in diffusional equilibrium (i.e. chemical composition of the system is uniform), we have to specify $(C - 1)$ composition variables, where C is the number of components in the system.

As mentioned earlier, a system is in thermodynamic equilibrium means that it is in mechanical equilibrium, thermal equilibrium and diffusional equilibrium. To specify these equilibria, we have to give p (i.e. mechanical variable), T (i.e. thermal variable) and $(C - 1)$ composition variables. So, there are $(C + 1)$ variables together. All these variables are intensive variables, i.e. do not depend on the size of the system.

1.2. Gibbs Phase Rule

When we specify the T , p , and $(C - 1)$ composition variables for a phase, the state of the phase is fully described. If the system contains a number of phases in equilibrium, then by specifying these variables for all the phases, we can describe the state of the system completely.

Let us compile equations expressing equilibrium conditions for the most general case, when a system consists of P number of phases, each of which includes all the components without any exception.

Thus, Number of phases = P;

Number of components = C.

Denoting the components of the system by the subscripts 1,2,3,.....C, and the phases by superscripts I, II, III,.....P, we can write the following equations for equilibrium, when there are C number of components and P number of phases.

$$\left. \begin{array}{l} T^I = T^{II} = T^{III} = \dots\dots\dots = T^P \\ p^I = p^{II} = p^{III} = \dots\dots\dots = p^P \end{array} \right\} \quad (1.3.a)$$

$$\left. \begin{array}{l} \mu_1^I = \mu_1^{II} = \mu_1^{III} = \dots\dots\dots = \mu_1^P \\ \mu_2^I = \mu_2^{II} = \mu_2^{III} = \dots\dots\dots = \mu_2^P \\ \mu_3^I = \mu_3^{II} = \mu_3^{III} = \dots\dots\dots = \mu_3^P \\ \dots\dots\dots \\ \mu_C^I = \mu_C^{II} = \mu_C^{III} = \dots\dots\dots = \mu_C^P \end{array} \right\} \quad (1.3.b)$$

The series of equations (1.3.a) are a series of identities, since the pressure and temperature are independent variables determining the state of the system.

The series of equations (1.3.b) are not a series of identities, since the chemical potential of the same component in different phases is described by different functions of concentrations (i.e. concentrations can vary from one phase to another), temperature and pressure. Independent equations can be formed on their basis.

Let us count the number of equations forming a system of independent equations composed on the basis of the series of equations (1.3.b), and the number of independent variables in these equations.

Each line of the system of equations (1.3.b) consist of $(P - 1)$ independent equations. The number of lines in the system is equal to C , and therefore the total number of independent equations is equal to $C(P - 1)$.

The T , p and concentrations of components are independent variables entering into the given system of equations. The number of independent concentrations (composition variables) in a phase is equal to $(C - 1)$ (when the components do not react with one another). So, the total number of composition variables in all P phases is $P(C - 1)$. In addition to the found number of concentrations, p and T are also independent variables. So, the total number of independent variables included in the system of equations is equal to $[P(C - 1) + 2]$.

If the number of independent variables is equal to the number of equations relating them,

$$\text{i.e. } C(P - 1) = P(C - 1) + 2 \quad (1.4),$$

then each independent variable acquires a certain definite value and the entire system can exist with this single possible combination of values of the temperature, pressure and the concentrations of the components in all phases.

If the number of independent equations is less than the number of independent variables, the difference, denoted by the symbol F ,

$$\text{i.e. } F = [P(C - 1) + 2] - [C(P - 1)] \quad (1.5)$$

is the number of variables which can be given arbitrary values with the available number of equations.

The quantity F , which is called the number of degrees of freedom, is the number of independent variables necessary to specify the system completely. F is also frequently called the variance of the system.

Thus a system having one degree of freedom is called a univariant one, a system with two degrees of freedom is called a bivariant one etc. If the number of degrees is equal to zero, the system is called an invariant one.

We can re-write the equation 1.5 as follows:

$$F = [P(C - 1) + 2] - [C(P - 1)] = PC - P + 2 - PC + C$$

$$F = C + 2 - P \quad (1.6)$$

This relationship, derived by J.W. Gibbs in 1876, is known as the Gibbs Phase Rule.

Gibbs phase rule can be formulated as follows:

“The number of degrees of freedom of an equilibrium thermodynamic system, which of all the external factors is influenced only by pressure and temperature is equal to the number of components of the system plus two, minus the number of phases.”

1.3. Thermodynamics of Phase Changes in Univariant Systems

In section 1.1, we have discussed a simple example of phase equilibrium, where water molecules in the liquid form represent one phase, and in the gaseous form represent another phase. Usually, the physical change of one state of matter (solid, liquid or gaseous) to another state of matter represents a phase change. Transformation of one allotropic form of an element to another (e.g. rhombic sulphur to monoclinic sulphur) is also a typical example of a phase change. Although, there is no change in the state of matter (both forms being solids), each allotropic form constitutes a separate phase.

A change of phase from one to another in a one component system ($C = 1$), constitutes a univariant system, and the specification of one independent variable determines the system completely. Phase changes in pure one component systems, such as water and allotropic changes are all examples of univariant systems in which the number of degrees of freedom (F) is one.

1.3.1. Entropy Change (ΔS) in Univariant Phase Changes

A univariant phase transformation taking place under equilibrium conditions, is always a reversible process, since the single degree of freedom possessed by such a system represents a unique condition for both directions of the phase change. For example, the melting of ice to water takes place under standard atmospheric pressure at the unique temperature of 273 K.

Liquid water freezes to ice under standard atmospheric pressure at the same temperature. It means that a reversible phase change occurs between one state of matter and another under the relevant conditions of T and p.

So, it follows that in a phase transformation under equilibrium conditions, the heat change takes place in a reversible manner under isobaric ($p = \text{const.}$) and isothermal ($T = \text{const.}$) conditions.

$$q = q_{\text{rev}} = \Delta H \quad (1.7)$$

According to the Second Law of Thermodynamics,

$$\Delta S = \int \delta q_{\text{rev}}/T$$

Since $T = \text{const.}$,

$$\Delta S = q_{\text{rev}}/T \quad (1.8)$$

For a univariant phase change, from equations (1.7) and (1.8), we can write

$$\Delta S = \Delta H/T \quad (1.9)$$

1.3.2. Gibbs free Energy Change (ΔG) in Univariant Phase Changes

We have learned that univariant phase transformations take place under the conditions of an equilibrium T and p. It follows that the Gibbs free energy criterion for equilibrium

$$dG_{p,T} = 0 \text{ or } \Delta G_{p,T} = 0 \quad (1.10),$$

should apply for all univariant phase transformations in equilibrium.

From equation (1.10), we can write

$$dG_1 = dG_2 \quad (1.11)$$

From the fundamental thermodynamic equations, we have

$$dG = VdP - SdT \quad (1.12)$$

$$dG_1 = V_1dP - S_1dT \quad (1.13)$$

$$dG_2 = V_2dP - S_2dT \quad (1.14)$$

(1.14) – (1.13), gives

$$dG_2 - dG_1 = (V_2 - V_1) dP - (S_2 - S_1) dT = 0$$

$$dP/dT = (S_2 - S_1)/(V_2 - V_1) = \Delta S/\Delta V \quad (1.15)$$

On other hand, the equation (1.9):

$$\Delta S = \Delta H/T$$

From equations (1.9) and (1.15), we have

$$dP/dT = \Delta H/T\Delta V \quad (1.16)$$

The equation (1.16) is known as the Clapeyron Equation, and expresses the variation of the equilibrium (vapour) pressure of a reversible phase transition (in a univariant system) as a function of the temperature, in terms of ΔH , ΔV and the transition temperature T . The transition temperature will be the melting point for an $S \rightleftharpoons L$ equilibrium, boiling point for an $L \rightleftharpoons V$ equilibrium, and sublimation point for an $S \rightleftharpoons V$ equilibrium.

1.3.3. Clausius-Clapeyron Equation (One phase is ideal gas)

According to the Clapeyron equation:

$$dP/dT = \Delta H/T\Delta V \quad (1.16)$$

$$\text{where volume change } \Delta V = V_2 - V_1 = V_{\text{Phase II}} - V_{\text{Phase I}} \quad (1.17)$$

If one phase is (e.g. Phase II) is considered as the vapour phase, then the volume of vapour phase is very much larger than that of the other phase,

$$\text{i.e. } V_{\text{vapour}} = V_{\text{Phase II}} \gg V_{\text{Phase I}}.$$

$$\text{Then, } \Delta V \approx V_{\text{Phase II}} = V_{\text{vapour}}.$$

If the vapour behaves as an ideal gas, then we can write:

$$\Delta V = V_{\text{vapour}} = nRT/p \quad (1.18)$$

From equations (1.16) and (1.18), it follows that:

$$dP/dT = \Delta H.p/T(nRT)$$

$$dp/dT = (\Delta H.p)/nRT^2$$

$$dp/p = \Delta H/nR \cdot dT/T^2 \quad (1.19)$$

By integrating, we have:

$$\ln p = 1/nR \int \Delta H (dT/T^2) \quad (1.20)$$

This is one form of the Clausius-Clapeyron Equation.

If we assume that, ΔH is independent of T , then we have;

$$\ln p = \Delta H /nR \int T^{-2} dT \quad (1.21)$$

$$\ln p = - \Delta H /nRT + \text{Const.} \quad (1.22)$$

This is another form of Clausius-Clapeyron Equation.

It is more desirable and useful to do the integration between suitable initial and final limits:

$$p_2 \int_{p_1} dp/p = \Delta H /nR \int_{T_1}^{T_2} T^{-2} dT$$

$$\ln p_2/p_1 = \Delta H/nR [1/T_1 - 1/T_2] \quad (1.23)$$

The Clausius-Clapeyron Equation can be applied for a reversible phase transformation in a univariant system, where one phase involves an ideal gaseous phase, and there is a large volume change accompanying the transition. Clausius-Clapeyron Equation in the forms of (1.22) and (1.23), requires the enthalpy of vaporization also to be independent of temperature.